

Data EDA Report

Generated from the actual project data files.

Source Data

Item	Value
Cleaned NASA rows	52,608
EMS rows	52,608
Start timestamp	2020-01-01 05:30:00
End timestamp	2026-01-01 04:30:00
Duplicate timestamps after cleaning	0
Missing cells after cleaning	0

Dataset Meaning

The raw source is NASA POWER hourly meteorological data. The EMS dataset extends it with physically constrained microgrid operating signals:

Signal	Meaning
ghi, dni, diffuse_irradiance	Solar resource inputs
temperature_c, wind_speed_mps, humidity_pct, precipitation_mm	Weather drivers
solar_kw	PV generation derived from irradiance and PV capacity
load_kw	realistic synthetic facility demand
tariff_inr_kwh	India-style time-of-use tariff
battery_soc_pct, battery_power_kw	BESS state and charge/discharge power

Key Findings

Finding	Value
Peak load	151.91 kW
95th percentile load	108.57 kW
Solar zero or near-zero hours	48.10%
Total simulated load	3,844,827.29 kWh-equivalent
Total PV generation potential	1,275,872.00 kWh-equivalent
Strongest temperature-load lag	0 hours
Correlation at strongest lag	0.6072

Statistical Summary

stat	ghi	dni	diffuse_irradiance	temperature_c	wind_speed_mps	humidity_pct	precipitation_mm	solar_kw	load_kw	tariff_
count	52608.000	52608.000	52608.000	52608.000	52608.000	52608.000	52608.000	52608.000	52608.000	52608
mean	216.967	128.985	104.025	26.836	3.238	73.753	3.572	24.252	73.084	5.200
std	294.485	206.295	131.098	4.976	1.633	21.557	11.199	32.760	20.702	2.236
min	0.000	0.000	0.000	13.430	0.030	11.170	0.000	0.000	26.383	2.600
1%	0.000	0.000	0.000	17.050	0.540	20.920	0.000	0.000	36.756	2.600
5%	0.000	0.000	0.000	19.180	1.100	30.600	0.000	0.000	42.101	2.600
25%	0.000	0.000	0.000	23.590	2.060	59.830	0.000	0.000	56.550	2.600
50%	7.190	0.000	9.400	26.180	2.910	79.440	0.030	0.818	72.019	5.600
75%	428.428	210.432	211.492	29.820	4.200	91.930	2.070	48.191	88.382	5.600
95%	826.312	595.809	354.860	36.340	6.260	98.670	18.686	91.615	108.575	9.200
99%	941.670	754.370	430.438	39.640	8.140	100.000	48.567	103.633	118.667	9.200
max	1043.150	1100.000	540.250	44.520	14.630	100.000	300.000	114.615	151.913	9.200

Tariff Distribution

tariff_inr_kwh	hours
2.600	17536.000
5.600	26304.000
9.200	8768.000

EDA Plots

Plot	File
Correlation heatmap	data/outputs/eda/eda_correlation_heatmap.png
Hourly solar/load/tariff profile	data/outputs/eda/eda_hourly_profile.png
Monthly seasonality	data/outputs/eda/eda_monthly_profile.png
Temperature-load lag correlation	data/outputs/eda/eda_temperature_load_lag.png
Load duration curve	data/outputs/eda/eda_load_duration_curve.png
Signal distributions	data/outputs/eda/eda_distributions.png
Recent battery/grid profile	data/outputs/eda/eda_battery_grid_recent.png

Operator Interpretation

The data supports the EMS design because the highest-value control window is evening: solar output drops toward zero while load and tariff rise. Battery dispatch is therefore economically useful when it pre-charges during low tariff periods or solar surplus and discharges during peak tariff import. The lag correlation analysis also supports sequence models because load response is not explained only by the current weather hour.